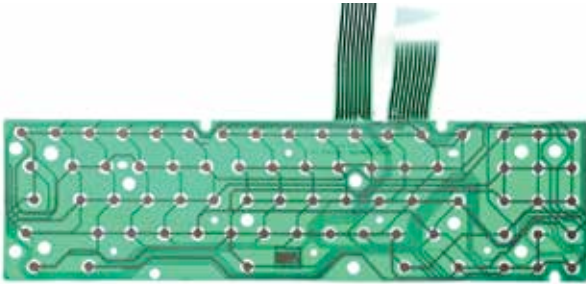


At the technology level the USB port uses polling to figure out if any particular port has anything to communicate. Whereas PS/2 is what we call an always on device. The way they both handle interrupts is different.

Key Roll Over

Interrupts are sort of handshakes which are used by devices to initialise a communication. Once an interrupt is sent by a hardware device the computer knows that it will start communicating soon and listens to



that device for data. On the USB, the interrupts are sent in packets, i.e. a cluster of data is sent to the computer and the computer has to break it down to figure out what key was pressed. The average person doesn't squeeze multiple keys at the same time. At most we press 3 keys which is what most operating system short-cut keys are limited to. Gamers on the other hand have the habit of mashing keys together, they develop key combinations which when executed in the right sequence could pretty much win the game for them, these sequences can sometimes be very rapidly executed. We know for a fact that multiplayer RTS games like StarCraft II have specific terminology for this, they call it APM or Actions Per Minute. A professional gamer has an APM upwards of 500-600, this equals to about 10 key presses per second and this is just the average. In heated moments this can scale much higher. Now keyboards were designed to handle around 3 simultaneous key presses, you could try this with your own computer, try pressing 4 keys which are in a horizontal line at the same time. If you do get it right then at most 2 keys should register (unless you are using a high end keyboard).

This is because the keyboard is unable to distinguish between multiple key presses and singles out what it can definitely say has been pressed. This is called KRO or Key Roll Over.